Clemson Means BUSINESS

Chapter 4
Structural Query Language (SQL)
Part II. Joining Tables

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Queries for Multiple Tables

- **Join:** A relational operation that causes two or more tables with a common domain to be combined into a single table or view

Order_T

OrderID	OrderDate	CustomerID
1001	10/21/2018	1
1002	10/21/2018	8
1003	10/22/2018	15
1004	10/22/2018	5
1005	10/24/2018	3
1006	10/24/2018	2
1007	10/27/2018	11
1008	10/30/2018	12
1009	11/5/2018	4
1010	11/5/2018	1
0		0

Record: 1 of 10 No Filter

Customer_T

CustomerID	CustomerName	CustomerAddress	CustomerCity	CustomerState	CustomerPostalCode
1	Contemporary Casuals	1355 S Hines Blvd	Gainesville	FL	32601-2871
2	Value Furniture	15145 S.W. 17th St.	Plano	TX	75094-7743
3	Home Furnishings	1900 Allard Ave.	Albany	NY	12209-1125
4	Eastern Furniture	1925 Beltline Rd.	Carteret	NJ	07008-3188
5	Impressions	5585 Westcott Ct.	Sacramento	CA	94206-4056
6	Furniture Gallery	325 Flatiron Dr.	Boulder	CO	80514-4432
7	Period Furniture	394 Rainbow Dr.	Seattle	WA	97954-5589
8	California Classics	816 Peach Rd.	Santa Clara	CA	96915-7754
9	M and H Casual Furniture	3709 First Street	Clearwater	FL	34620-2314
10	Seminole Interiors	2400 Rocky Point Dr.	Seminole	FL	34646-4423
11	American Euro Lifestyles	2424 Missouri Ave N	Prospect Park	NJ	07508-5621
12	Battle Creek Furniture	345 Capitol Ave. SW	Battle Creek	MI	49015-3401
13	Heritage Furnishings	66789 College Ave.	Carlisle	PA	17013-8834
14	Kaneohe Homes	112 Kiowai St.	Kaneohe	HI	96744-2537
15	Mountain Scenes	4132 Main Street	Ogden	UT	84403-4432

Record: 1 of 15 No Filter Search

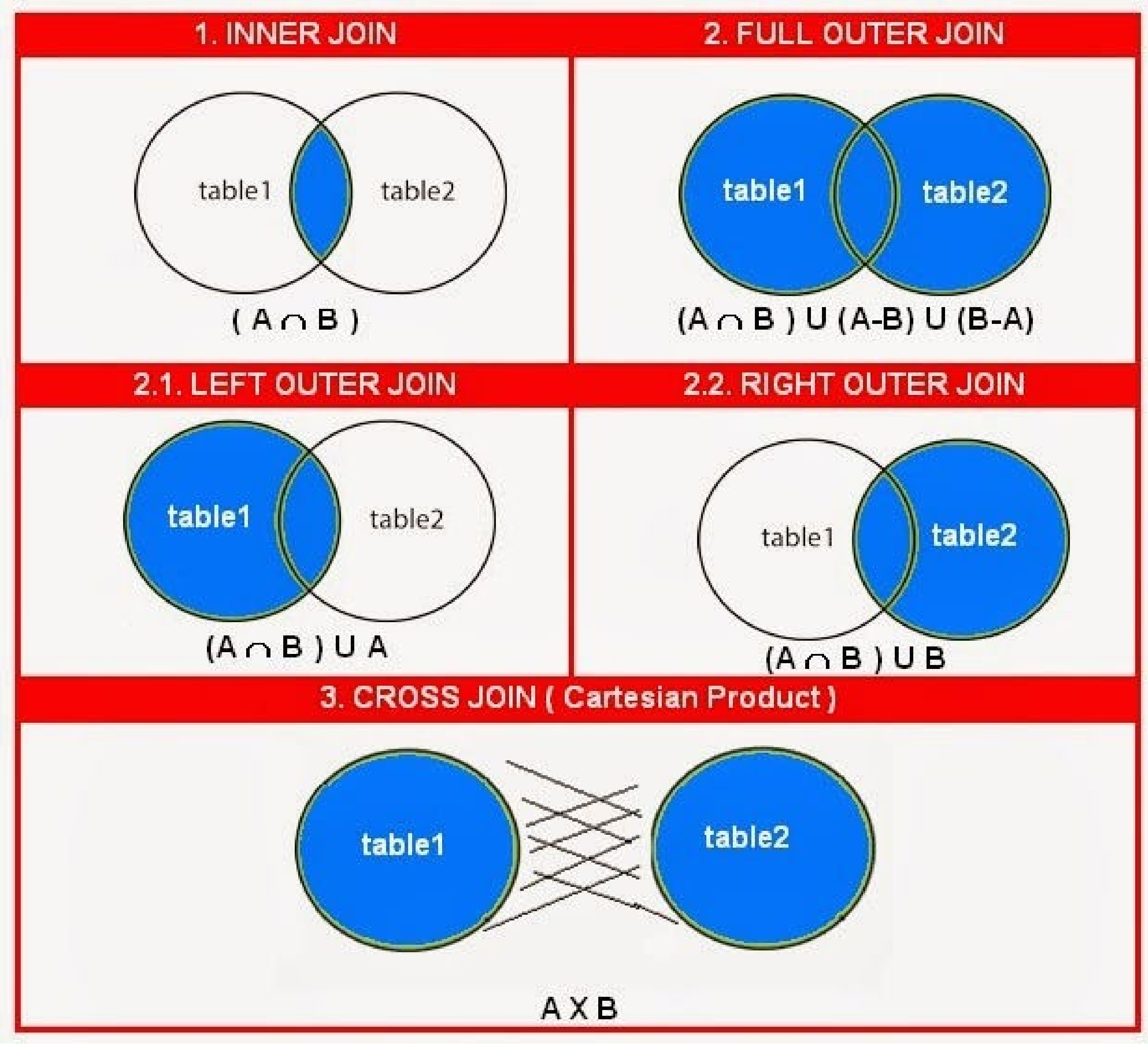
Types of Joins

Syntax: specify as **FROM** clause

table 1 explicit **JOIN** *table 2* **ON** joining condition

explicit: could be INNER, LEFT OUTER, RIGHT OUTER, FULL OUTER, or CROSS

With cross join, no need to specify condition



Inner Join

- **Q23:** What are the customer IDs and names of all customers, along with the order IDs for all the orders they have placed?

```
SELECT Customer_T.CustomerID, CustomerName, OrderID
FROM "db_pvfc11_big"."Customer_T" INNER JOIN Order_T
on Customer_T.CustomerID = Order_T.CustomerID;

SELECT Customer_T.CustomerID, CustomerName, OrderID
FROM "db_pvfc11_big"."Customer_T”, Order_T
WHERE Customer_T.CustomerID = Order_T.CustomerID;
```

CUSTOME...	CUSTOME...	ORDERID
16	ABC Furniture C	51
3	Home Furnishing	2
12	Flanigan Furnitur	47
9	A Carpet	66
14	Wild Bills	58
8	Dunkins Furniture	35
13	Ikards	59
6	Furniture Gallery	9
1	Contemporary C	46

Left Outer Join

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- **Q24:** List customer name, identification number, and order number for all customers listed in the customer table. Include the customer identification number and name even if there is no order available for that customer.

```
SELECT Customer_T.CustomerID, CustomerName, OrderID
FROM "db_pvfc11_big"."Customer_T" LEFT OUTER JOIN Order_T
on Customer_T.CustomerID = Order_T.CustomerID;
```

CUSTOME...	CUSTOME...	ORDERID
16	ABC Furniture Co	53
3	Home Furnishing	2
12	Flanigan Furnitur	45
9	A Carpet	66
5	Impressions	
14	Wild Bills	58
8	Dunkins Furniture	35
2	Value Furnitures	
13	Ikards	59
6	Furniture Gallery	44
1	Contemporary Ca	48
15	Janets Collection	34
4	Eastern Furniture	30
16	ABC Furniture Co	55

Page 1 of 2 (61 rows total)

Right Outer Join

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- **Q25:** List customer name, identification number, and order number for all customers listed in the customer table. Include the order number, even if there is no customer name and identification number available.

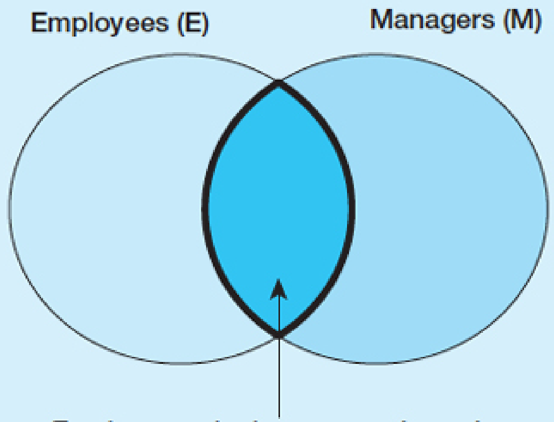
```
SELECT Customer_T.CustomerID, CustomerName, OrderID
FROM "db_pvfc11_big"."Customer_T" RIGHT OUTER JOIN Order_T
on Customer_T.CustomerID = Order_T.CustomerID;
```

CUSTOME...	CUSTOME...	ORDERID
16	ABC Furniture Co	52
3	Home Furnishing	2
12	Flanigan Furnitur	45
9	A Carpet	56
14	Wild Bills	58
8	Dunkins Furniture	50
13	Ikards	59
6	Furniture Gallery	44
1	Contemporary Ca	3
15	Janets Collection	34
4	Eastern Furniture	49
16	ABC Furniture Co	55
12	Flanigan Furnitur	47
9	A Carpet	66

Page 1 of 2 (58 rows total)

Self-Join

- A join requires matching rows in a table with other rows in that same table, e.g., unary relationship.



Employees who have supervisors; i.e.,
WHERE E.EmployeeSupervisor = M.EmployeeID

Employees (E)

EmployeeID	EmployeeName	EmployeeSupervisor
098-23-456	Sue Miller	
107-55-789	Stan Getz	
123-44-347	Jim Jason	678-44-546
547-33-243	Bill Blass	
678-44-546	Robert Lewis	

Managers (M)

EmployeeID	EmployeeName	EmployeeSupervisor
098-23-456	Sue Miller	
107-55-789	Stan Getz	
123-44-347	Jim Jason	678-44-546
547-33-243	Bill Blass	
678-44-546	Robert Lewis	

Self-Join

- **Q26:** What are the employee ID and name of each employee and the name of his or her supervisor?

```
SELECT E.EmployeeID, E.EmployeeName, M.EmployeeName AS Manager
FROM "db_pvfc11_big"."Employee_T" AS E, Employee_T AS M
WHERE E.EmployeeSupervisor = M.EmployeeID;
```

EMPLOYEEID	EMPLOYEE...	MANAGER
555955585	Mary Smith	Lawrence Haley
454-56-768	Robert Lewis	Phil Morris
Laura	Laura Ellenburg	Robert Lewis
123-44-345	Phil Morris	Robert Lewis
332445667	Lawrence Haley	Robert Lewis

5 rows total

Subqueries

- Placing an inner query (SELECT … FROM … WHERE) within a WHERE or HAVING clause of another query
- **Joining Technique:** when data from several relations are to be retrieved and displayed and the relationships are not necessarily nested;
- **Subquery Technique:** display data from only the tables mentioned in the outer query

Subqueries

- **Q27:** What are the name and address of the customer who placed order number 4?

Joining Technique:

```
SELECT CustomerName, CustomerAddress, CustomerCity, CustomerState,
CustomerPostalCode
FROM "db_pvfc11_big"."Customer_T", Order_T
WHERE Customer_T.CustomerID = Order_T.CustomerID
   AND OrderID = 4;
```

Subquery Technique:

```
SELECT CustomerName, CustomerAddress, CustomerCity, CustomerState,
CustomerPostalCode
FROM "db_pvfc11_big"."Customer_T"
WHERE CustomerID =
   (SELECT Order_T.CustomerID
    FROM "db_pvfc11_big"."Order_T"
    WHERE OrderID = 4);
```

Subqueries

- **Q28:** What are the names of customers who have placed orders?

```
SELECT CustomerName
FROM "db_pvfc11_big"."Customer_T"
WHERE CustomerID IN
  (SELECT DISTINCT CustomerID
   FROM "db_pvfc11_big"."Order_T");
```

CUSTOMERNAME
ABC Furniture Co.
Home Furnishings
Flanigan Furniture
A Carpet
Wild Bills
Dunkins Furniture
Ikards
Furniture Gallery
Contemporary Casuals
Janets Collection
Eastern Furniture

Subqueries

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- The qualifiers NOT, ANY, and ALL may be used in front of IN or with logical operators such as =, >, and <.
- Also can use < ANY or >= ALL

- **Q29:** Which customers have not placed any orders for computer desk?

```
SELECT CustomerName
FROM "db_pvfc11_big"."Customer_T"
WHERE CustomerID NOT IN
  (SELECT DISTINCT CustomerID
   FROM "db_pvfc11_big". "Order_T", "db_pvfc11_big".
"OrderLine_T", "db_pvfc11_big". "Product_T"
   WHERE Order_T.OrderID = OrderLine_T.OrderID
      AND OrderLine_T.ProductID = Product_T.ProductID
      AND ProductDescription = 'Computer Desk');
```

Subqueries

- Also can use ANY or ALL

- **Q30:** List the details about the product with the highest standard price.

```
SELECT ProductDescription, ProductFinish, ProductStandardPrice
FROM "db_pvfc11_big". "Product_T"
WHERE ProductStandardPrice >= ALL
  (SELECT ProductStandardPrice
   FROM "db_pvfc11_big". "Product_T");
```

EXISTS, NOT EXISTS

- EXISTS: True if the subquery returns an intermediate results table that is not empty; False if the subquery returns an empty table.
- NOT EXISTS: reverse of EXISTS

- **Q31:** What are the order IDs for all orders that have included furniture finished in Oak?

```
SELECT DISTINCT OrderID
FROM "db_pvfc11_big". "OrderLine_T"
WHERE EXISTS
  (SELECT *
   FROM "db_pvfc11_big"."Product_T"
   WHERE ProductID = OrderLine_T.ProductID
     AND ProductFinish = ‘Oak');
```